AP CALCULUS AB

Unit 1 • Worksheet 1-8A

Worksheet 1-8A • MVT

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What does the MVT need on $[a, b]$?

(A) f continuous on $[a, b]$ and differentiable on (a, b)

(B) f increasing

(C) $f(a) = f(b)$

(D) f continuous on (a, b) only

Question 2

NO CALCULATOR

f is differentiable, $f(0) = -2$ and $f(4) = 10$. The MVT guarantees a c in $(0, 4)$ with $f'(c) =$

(A) 12

(B) 3

(C) 2

(D) $\frac{1}{3}$

Question 3

NO CALCULATOR

For which function does the MVT apply on $[0, 2]$?

(A) $|x - 1|$

(B) $\frac{1}{x-1}$

(C) $x^2 - 3x$

(D) $\begin{cases} x, & x < 1 \\ x + 1, & x \geq 1 \end{cases}$

Question 4

NO CALCULATOR

h is differentiable.

x	1	4	6
$h(x)$	2	11	5

The MVT guarantees a c in $(4, 6)$ with $h'(c) =$

(A) 3

(B) 6

(C) -6

(D) -3

Question 5

NO CALCULATOR

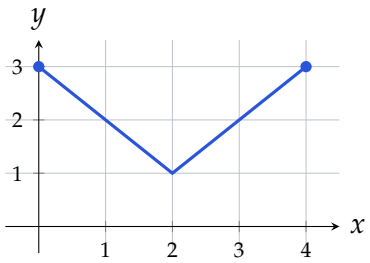
f is differentiable and $f(1) = f(5) = 3$. What does the MVT guarantee?

(A) A c in $(1,5)$ with $f'(c) = 0$ (a horizontal tangent)
(B) $f(3) = 3$
(C) f is constant.
(D) A c in $(1,5)$ with $f(c) = 0$

Question 6

NO CALCULATOR

The graph of $g(x) = |x-2| + 1$ on $[0,4]$ is shown. Does the MVT guarantee a c in $(0,4)$ with $g'(c) = \frac{g(4)-g(0)}{4-0}$?



(A) Yes, $c = 2$.
(B) No: g is not differentiable at $x = 2$.
(C) Yes, because g is continuous.
(D) No, because $g(0) = g(4)$.

Question 7

NO CALCULATOR

In words, the MVT says that somewhere the slope of a _____ equals the slope of the _____.

(A) secant line; tangent line
(B) tangent line; x-axis
(C) tangent line; secant line
(D) graph; table

Question 8 (Free response)

NO CALCULATOR

The water level $W(t)$, in meters, is differentiable.

t (h)	0	4	6	10
$W(t)$	2	5	9	7

(a) Explain why there must be a time t in $(4,6)$ with $W'(t) = 2$.
(b) Must there be a time in $(0,10)$ when the level rises at 0.5 m/h? Justify.
(c) Does the MVT apply to $g(x) = |x-3| + 1$ on $[1,5]$? Explain.